

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL

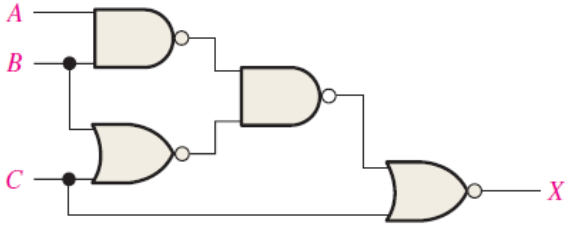
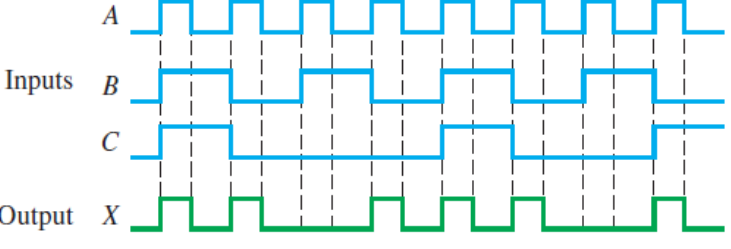
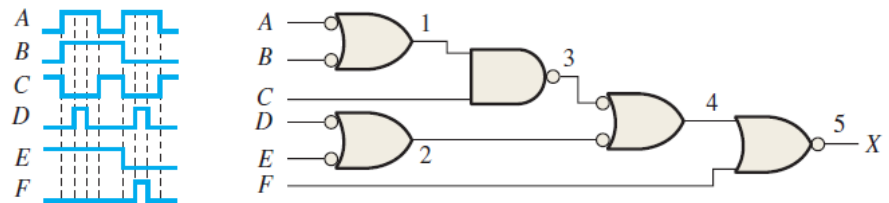
Department of Electrical Engineering

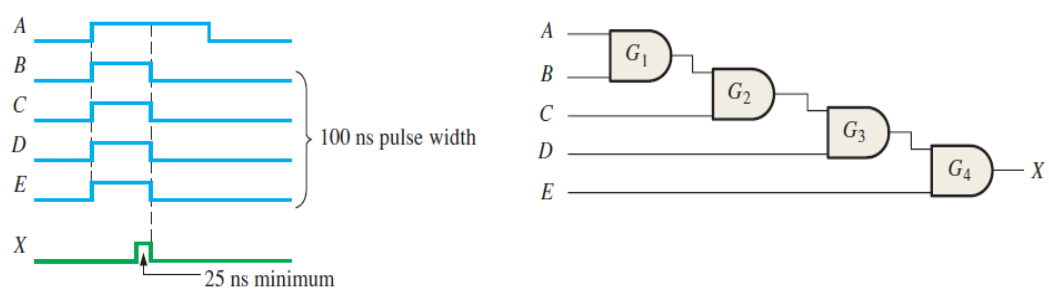
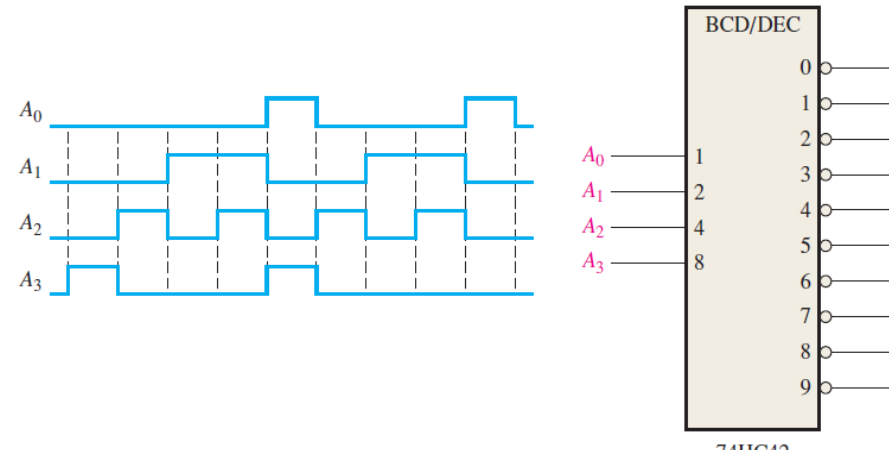
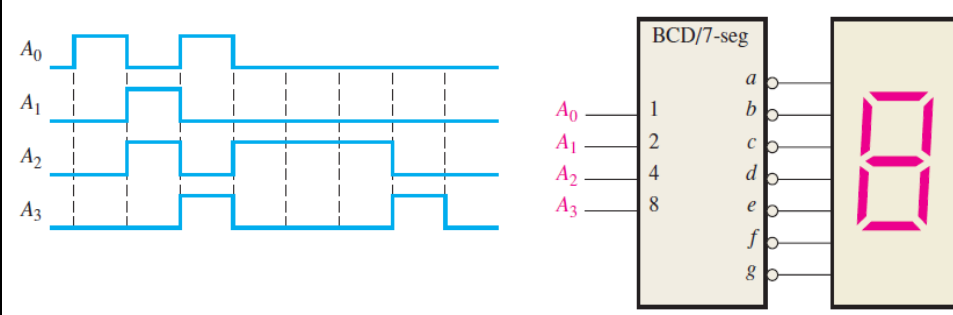
B. Tech. IV Semester

Sub: Electronics Devices & Circuits (EE-223)

Assignment – 4

Submission deadline: 21.04.2025

Q.1.	Develop an AND-OR-Invert logic circuit for a power drive which switches on (logic 1) when the guard is in place (logic 1) and switches off (logic 0) when the motor is too hot (logic 0).
Q.2.	Minimize the gates required to implement the function of the circuit in each part of Figure 1 in SOP form.  Fig. 1
Q.3.	The output of the logic circuit and input waveforms in Figure 2 is passed through an inverter. Draw the output waveform.  Fig. 2
Q.4.	For the input waveforms in Figure 3, what logic circuit will generate the output waveform shown?  Fig. 3
Q.5.	For the circuit in Figure 4, draw the waveforms at the numbered points in the proper relationship to each other.  Fig. 4

Q.6.	Assuming a propagation delay through each gate of 10 nanoseconds (ns), determine if the desired output waveform X in Figure 5 (a pulse with a minimum $t_w = 25$ ns positioned as shown) will be generated properly with the given inputs.  Fig. 5
Q.7.	You wish to detect only the presence of the codes 1010, 1100, 0001, and 1011. An active-HIGH output is required to indicate their presence. Develop the minimum decoding logic with a single output that will indicate when any one of these codes is on the inputs. For any other code, the output must be LOW
Q.8.	BCD numbers are applied sequentially to the BCD-to-decimal decoder in Figure 6. Draw a timing diagram, showing each output in the proper relationship with the others and with the inputs.  Fig. 6
Q.9.	A 7-segment decoder/driver drives the display in Figure 7. If the waveforms are applied as indicated, determine the sequence of digits that appears on the display.  Fig. 7
Q.10	A 74HC147 encoder has LOW levels on pins 2, 5, and 12. What BCD code appears on the outputs if all the other inputs are HIGH?

Q.11	For the multiplexer in Figure 7, determine the output for the following input states: $D_0 = 1$, $D_1 = 0$, $D_2 = 0$, $D_3 = 1$, $S_0 = 0$, $S_1 = 1$. Fig. 7
Q.12	The waveforms in Figure 8 are observed on the inputs of a 74HC151 8-input multiplexer. Sketch the Y output waveform Fig. 8
Q.13	Develop the total timing diagram (inputs and outputs) for a 74HC154 used in a demultiplexing application in which the inputs are as follows: The data-select inputs are repetitively sequenced through a straight binary count beginning with 0000, and the data input is a serial data stream carrying BCD data representing the decimal number 2468. The least significant digit (8) is first in the sequence, with its LSB first, and it should appear in the first 4-bit positions of the output.
Q.14	If the input waveforms are applied to the decoding logic as indicated in Figure 9, sketch the output waveform in proper relation to the inputs. Fig. 9

Q.15

When a LOW is on the output of each of the decoding gates in Figure 10, what is the binary code appearing on the inputs? The MSB is A_3 .

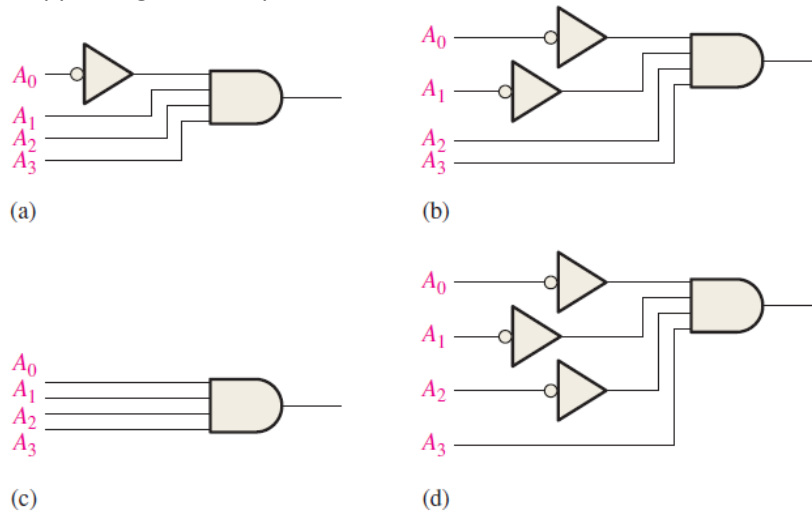


Fig. 10